Roma Tre

Information Visualization

Data modeling

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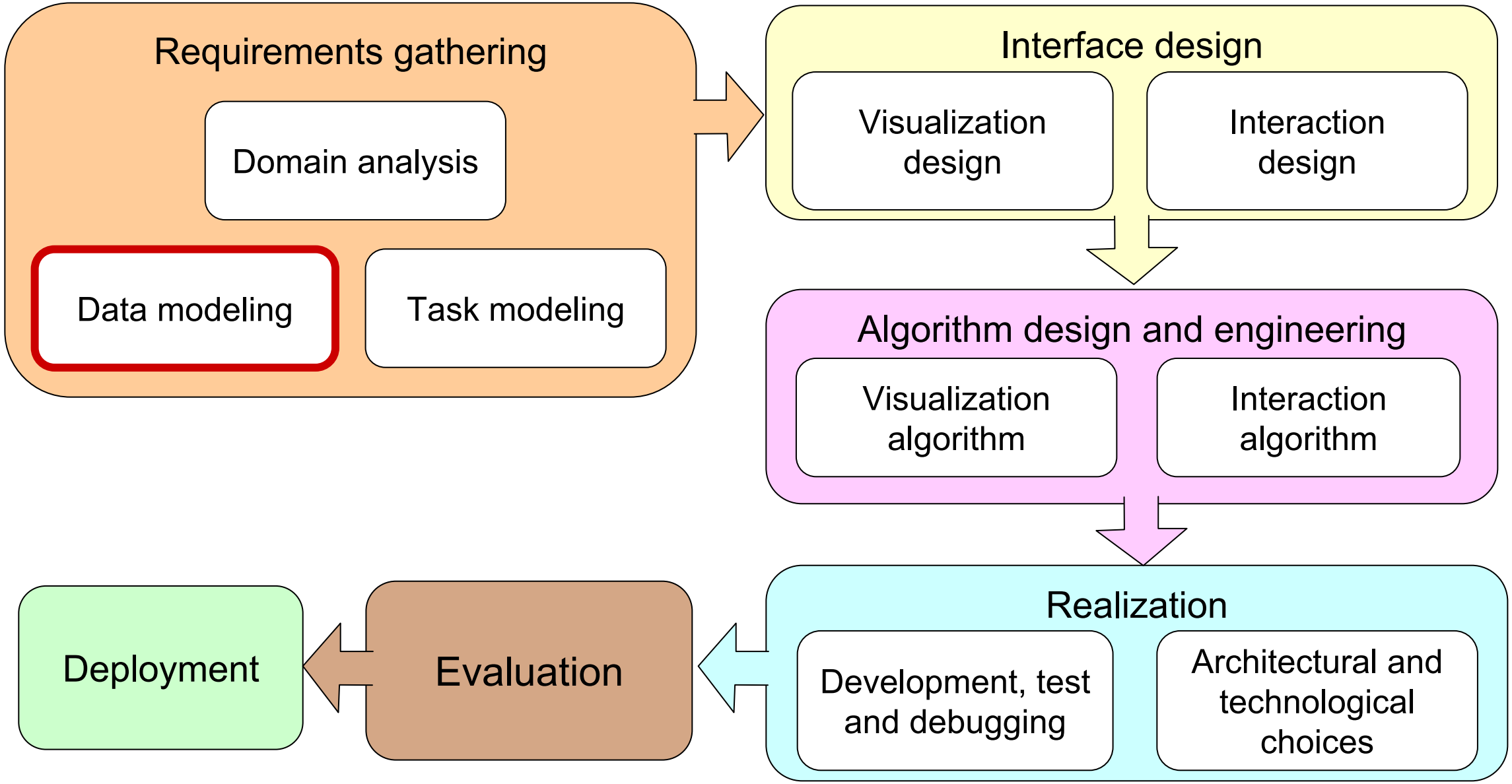
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Summary

- Data and data types
- Data cleaning and data transformation
- Modeling data tables and relationships
- Types of values
- Using data types to classify visualization problems and solutions

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Design pipeline



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Data

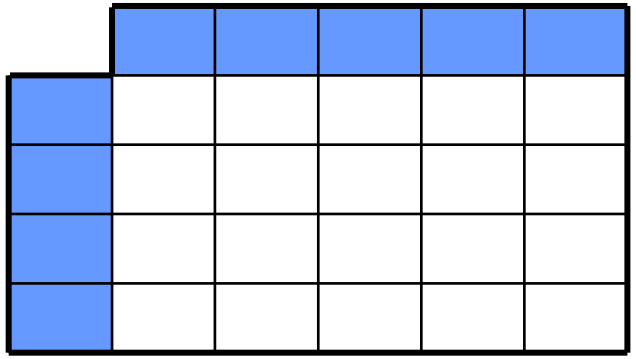
- Information is obtained from data
- In this lecture we focus on the latter
  - data represent some phenomenon or fact
  - data describe something of interest
- A definition for “data”
  - “data are values of qualitative or quantitative variables, belonging to a set of items”
    - http://en.wikipedia.org/wiki/Data
- Usually real world data don’t come in the form we would like them to be



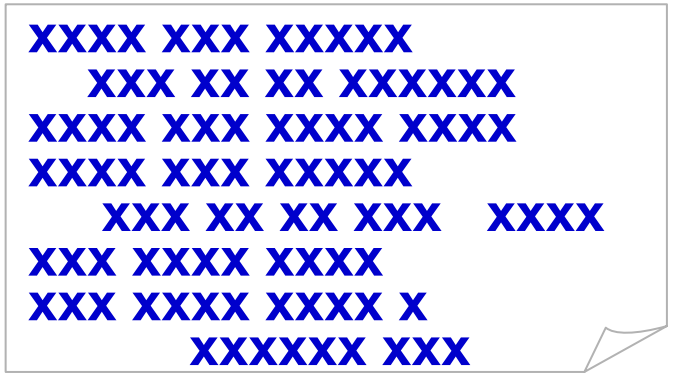
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The nature of real-world data

- Structured data
  - estimated 20% of the whole
  - defined length and format
  - examples
    - names, dates, integer or real values



- Unstructured data
  - estimated 80% of the whole
  - no fixed format
  - examples
    - text, streams, tweets, blogs

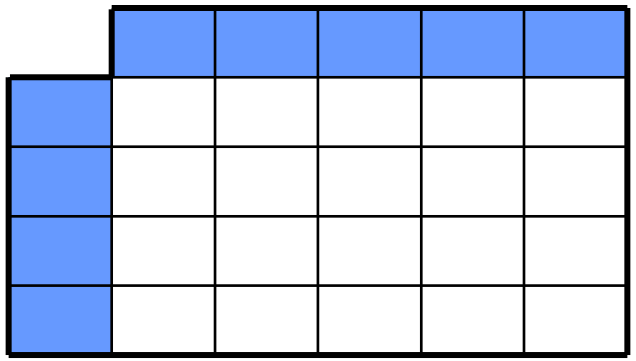


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Examples of structured data

- Computer-generated
  - sensor data
  - web log data
  - point-of-sale data
  - financial data
  - …
- Human-generated
  - input data
  - click stream data
  - gaming related data
  - …

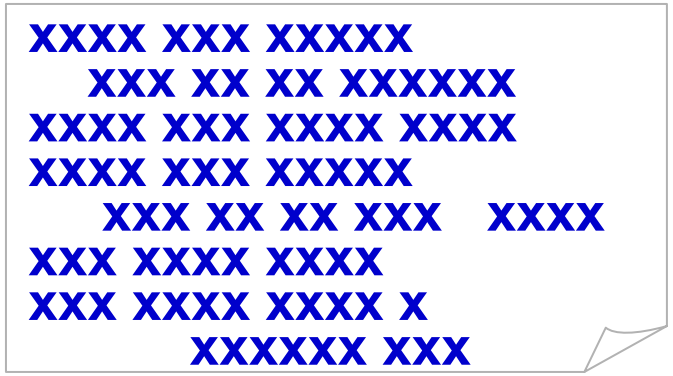
- All these data are in some structured format (e.g., tables)



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Examples of unstructured data

- Computer-generated
  - satellite images
  - scientific data
  - photographs and video
  - radar or sonar data
  - …
- Human-generated
  - text
  - social media data
  - website content
  - …
- All these data need to be analyzed to extract entities and values



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Semi-structured data

- Sometimes, it is convenient to distinguish a third class of data which is intermediate between structured and unstructured
- Semi-structured data
  - does not necessarily conform to a fixed format
  - may be self-decribing
  - example
    - simple label/value pairs

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Examples of semi-structured data

- Computer-generated
  - EDI
    - Electronic Data Interchange
  - SWIFT
    - Society for Worldwide Interbank Financial Telecommunication
  - XML-like
  - JSON
- Human-generated
  - organized text
    - Last name: Jones, First name: Jane, and Telephone: 348-4812633

EDI example

```
HDR|6||||
DTL|1|ABC|xyz|123|1
DTL|13|ABC|animal|334|1
DTL|11|ABC|sfdk|432|2
DTL|12|ABC|wewdc|3|1
DTL|14|ABC|qwdx|416|4
```

SWIFT example

```
MT542 DELIVER FREE
:16R:GENL
:20C::SEME//YOUR REF
:23G:NEWM
:16S:GENL
:16R:TRADDET
```


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The data modeling process

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Data pipeline

- Data extraction
  - transform all available data into structured data
- Data cleaning
  - reconstruct the data as if it were obtained from an ideal source
- Data transformation
  - model the data so that it is more useful for the user
- Data visualization

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Data extraction

transform all available data into structured data

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Data extraction

- Data gathering
  - download data files
  - harvest or scrape the Web
  - collect messages from social media
  - ...
- Producing structured data
  - convert file formats
  - parse text
  - recognize named entities (NER)
    - here «entity» is same as «item»
  - perform sentiment analysis
  - ...

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Data format conversion

- Needed to handle heterogeneous sources
- Example of data converson tool
  - Mr. Data Converter
    - created by Shan Carter (NYT)
    - written in javascript
    - works in your browser
    - convert tables from CSV to HTML, XML, JSON, MySQL, etc.
    - http://shancarter.github.io/mr-data-converter/

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Example of named-entity recognition

- Extracting names from raw text
  - Mr. People
    - created by Matthew Ericson (NYT)
    - identifies and extracts the first and last names, along with middle initial, prefix, and suffix
    - http://people.ericson.net/ (no longer available)
  - MonkeyLearn
    - https://monkeylearn.com/name-extractor-online/
- Transforms unstructured text into structured data

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Output of the data extraction process

- Three components have to be identified in order to get structured data
  - objects
    - entities, items of interest
  - attributes
    - properties, characteristics, values
  - relationships
    - describe how objects relate to each other
    - more on these later

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Table example

- Each row is a record or, better, a case

	SSN	Age	Hair	Grade	…
Mary	145	23	brown	B	…
Jim	294	17	black	A	…
Sally	563	47	blonde	B	…
Mitch	823	37	red	C	…
…	…	…	…	…	…

From [Stasko ‘13]

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Table example

- Each row is a record or, better, a case

Name	SSN	Age	Hair	Grade	…
Mary	145	23	brown	B	…
Jim	294	17	black	A	…
Sally	563	47	blonde	B	…
Mitch	823	37	red	C	…
	…	…	…	…	…

Do not allow cases to have metadata attached to them

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Variable types

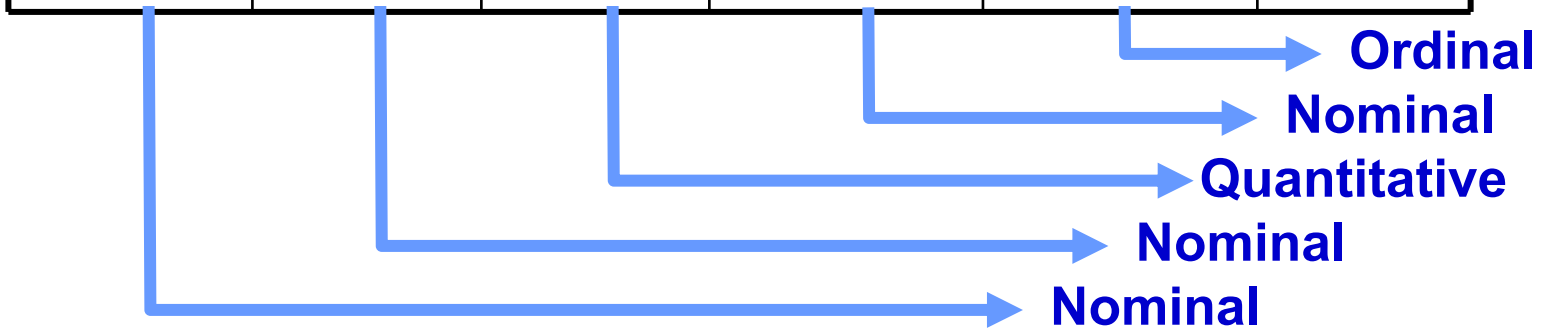
- Quantitative
  - ordered, measurable
  - examples
    - integer or reals
    - time
- Ordinal (labels expressing degree)
  - qualitative, ordered, but not at measurable intervals
  - examples
    - first, second, third, …
    - cold, warm, hot
- Nominal or categorical (labels)
  - qualitative, no inherent order, not necessarily unique
  - examples
    - gender, city names, types of diseases, …

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Table example

- Classify variable types

Name	SSN	Age	Hair	Grade	…
Mary	145	23	brown	B	…
Jim	294	17	black	A	…
Sally	563	47	blonde	B	…
Mitch	823	37	red	C	…
	…	…	…	…	…



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Data cleaning

reconstruct the data as if it were obtained from an ideal source

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Data cleaning

- Data may be inconsistent and messy
  - some values may be missing or corrupted
  - figures may be expressed with different scales
    - hundreds, thousands?
  - several strings may be equivalent
    - “R&D”, “r&d”, “Research and Development”, “RD”, “R. and D.”, etc.
- We would like to have tidy, structured data

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Data cleaning

- Transforms raw data into tidy, structured data
- Purposes
  - remove errors
  - simplify the data
- Output
  - clean and structured data
  - metadata (sometimes called “code book”) that describes exactly what the values are
  - a precise description of the performed transformations
- May be pursued with different approaches
  - human guided (“by inspection”)
  - automatic (“by scripts”)

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Metadata

- Descriptive information about the data
  - specifies the role of the variables
    - example: variable1 is “student grading”
  - specifies the type of the variables
    - example: variable1 is a single character
  - specifies constraints
    - example: if variable1 is “A”, then variable3 can only be 3, 7 or 16

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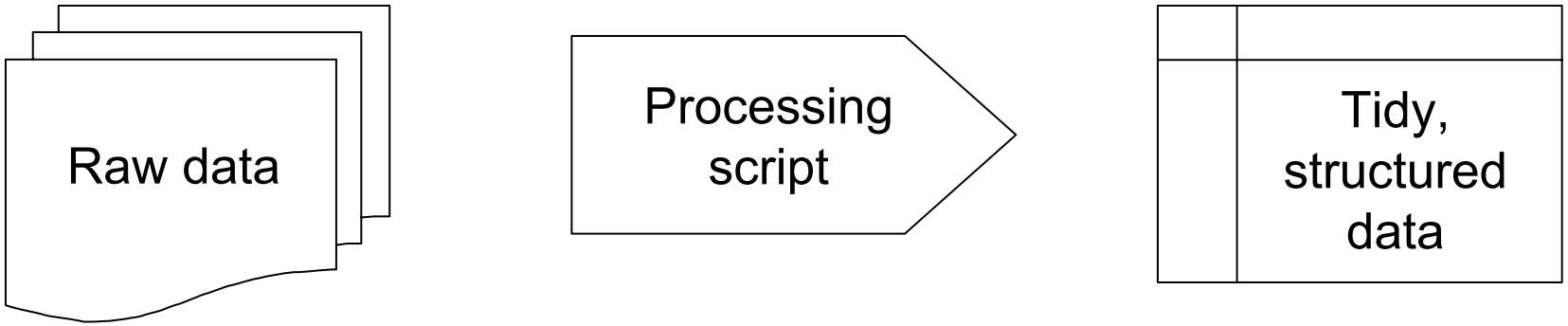
Human guided data refinement

- Refining by inspection and macro-operations
  - OpenRefine (ex-Google Refine)
    - desktop application (no need to upload data)
    - spreadsheet-like interface
    - allows to count, compare, and automatically cluster string and numeric values
    - allows to reassign values based on similarities
    - see the first video at http://openrefine.org/ to have an idea

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Automatic data refinement

- Ideally, we would like to have the following pipeline



- All steps of the transformation pipeline should be recorded
  - to make sure that no artifact is introduced by the transformation pipeline
  - to allow for reproducible analysis
- The script should have no parameters
  - all parameters are embedded in the script

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How raw can you go?

- Rule of thumb
  - start from data in the most raw format you can handle
    - since you are yourself taking care of the transformation pipeline
- Double check that
  - you didn’t run software on the data
  - you didn’t change values
  - you didn’t remove any data from the data set
  - you didn’t summarize data in any way

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Data transformation

model the data so that it is more useful for the user

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Data transformation and modeling

- Once a clean dataset is obtained the data can be transformed in order to better satisfy user demands
- Common transformations include
  - data adjustment
    - change the data values
  - data simplification
    - reduce the quantity of data
  - data mapping
    - transform the data structure

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Data adjustment

- Math transformations
  - replace values with inverse
    - use 1/x instead of x
  - replace absolute values with percentages
    - with respect to the sum of all values
  - round and truncate data
    - example: cut out decimals
  - normalize values
    - scale between 0 and 1, where 1 corresponds to the maximum value
- Change variable type
  - map nominal to ordinal or quantitative
  - map quantitative to ordinal
  - convert format
    - example: timestamps to text dates

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Data simplification

- Filter or select
  - removes part of the data items
  - example: remove items containing null values
- Project
  - removes part of the variables
  - example: keep sender and recipient of a whole message
- Aggregate data
  - collapses several cases into one
  - replaces values with average, maximum, minimum, sum, count

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Data mapping

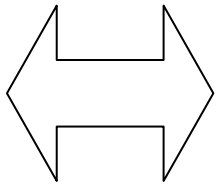
- Change the way the tables are organized
  - perform a join operation between two tables
  - embed metadata into the tables
  - move values into metadata
  - ...

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An example of data mapping

- Consider a table containing quotations from the stock market
- You can regard stock names as
  - values for the variable “Stock”
  - variable names (i.e., metadata)

Time	Stock	$
145	SDI	23
145	QQQ	17
146	LUM	47
146	SDI	24
…	…	…



Time	SDI	QQQ	LUM
145	23	17	-
146	24	-	47
147	22	18	47
148	23	-	50
…	…	…	…

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An example of data mapping

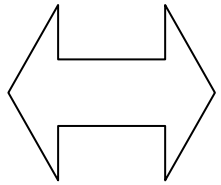
- Consider tables containing quotations from different days
- You can embed the metadata into the tables
  - for example, extracting data from the file naming convention

day2.xls

Time	Stock	$
		23
		17
		47
		24
		…

day1.xls

Time	Stock	$
145	SDI	23
145	QQQ	17
146	LUM	47
146	SDI	24
…	…	…



Day	Time	Stock	$
1	145	SDI	-
2	145	QQQ	47
1	146	QQQ	47
2	148	SDI	50
…	…	…	…

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Example of data modeling

- Dataset of Titanic passengers
  - from www.encyclopedia-titanica.org

Name	Age	Class	Ticket	boat [body]
ALLEN, Miss Elisabeth Walton	29	1°	24160	2
ANDERSON, Mr Harry	47	1°	19952	3
BATEMAN, Revd Robert James	51	2°	1166	[174]
BEAUCHAMP, Mr Henry James	28	2°	244358	
ADAMS, Mr John	26	3°	341826	[103]
…	…	…	…	…

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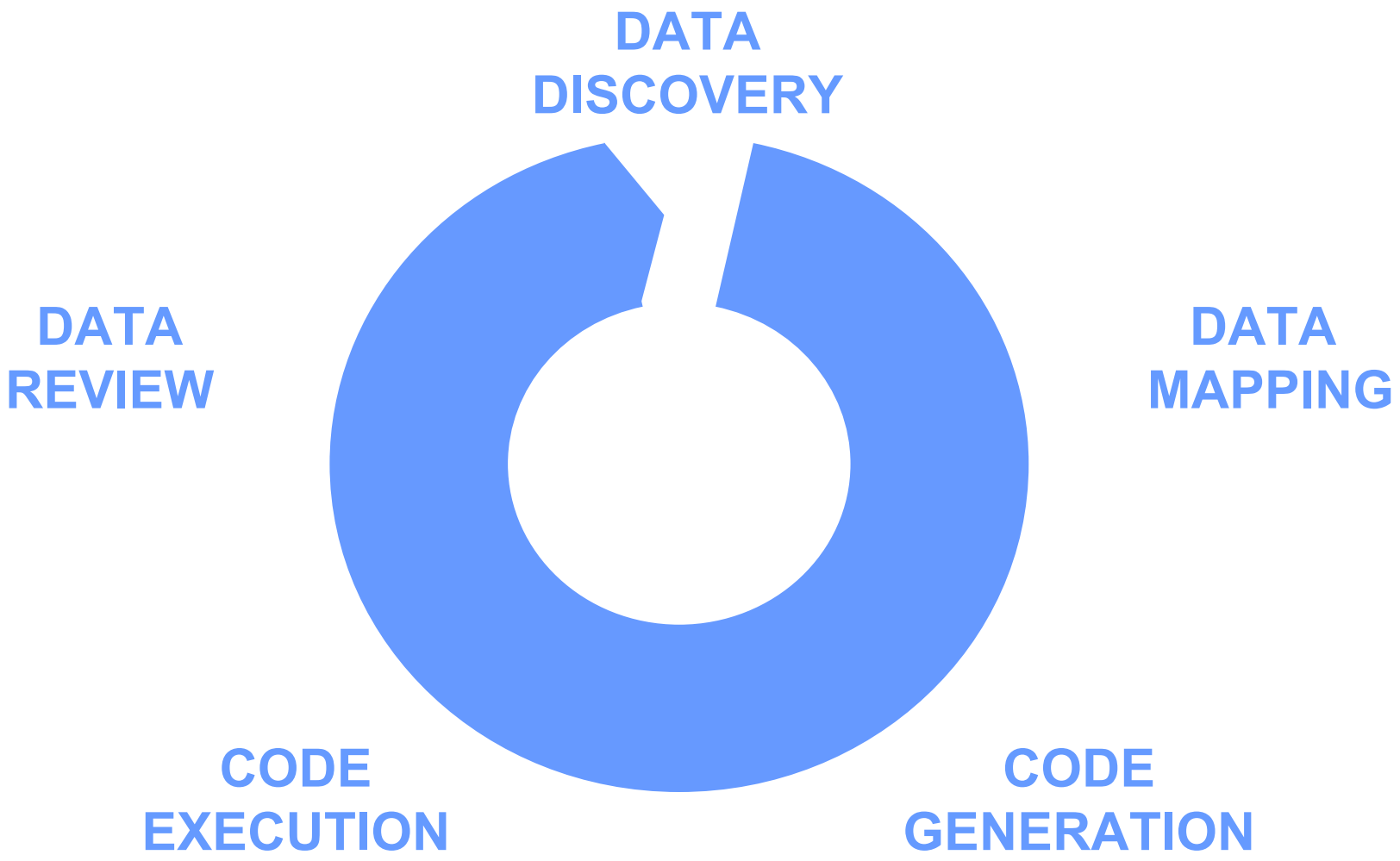
Example of data modeling

- Data transformation
  - parsing
    - extract gender
    - infer if survived or not
  - simplification
    - remove name and age
    - count by class

Gender	Survived	1° class	2° class	3° class	Crew
Male	No	118	154	422	670
Male	Yes	62	25	88	192
Female	No	4	13	106	3
Female	Yes	141	93	90	20

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Automatic data transformation



for example, https://adeptia.com/blog/data-transformation-steps-benefits

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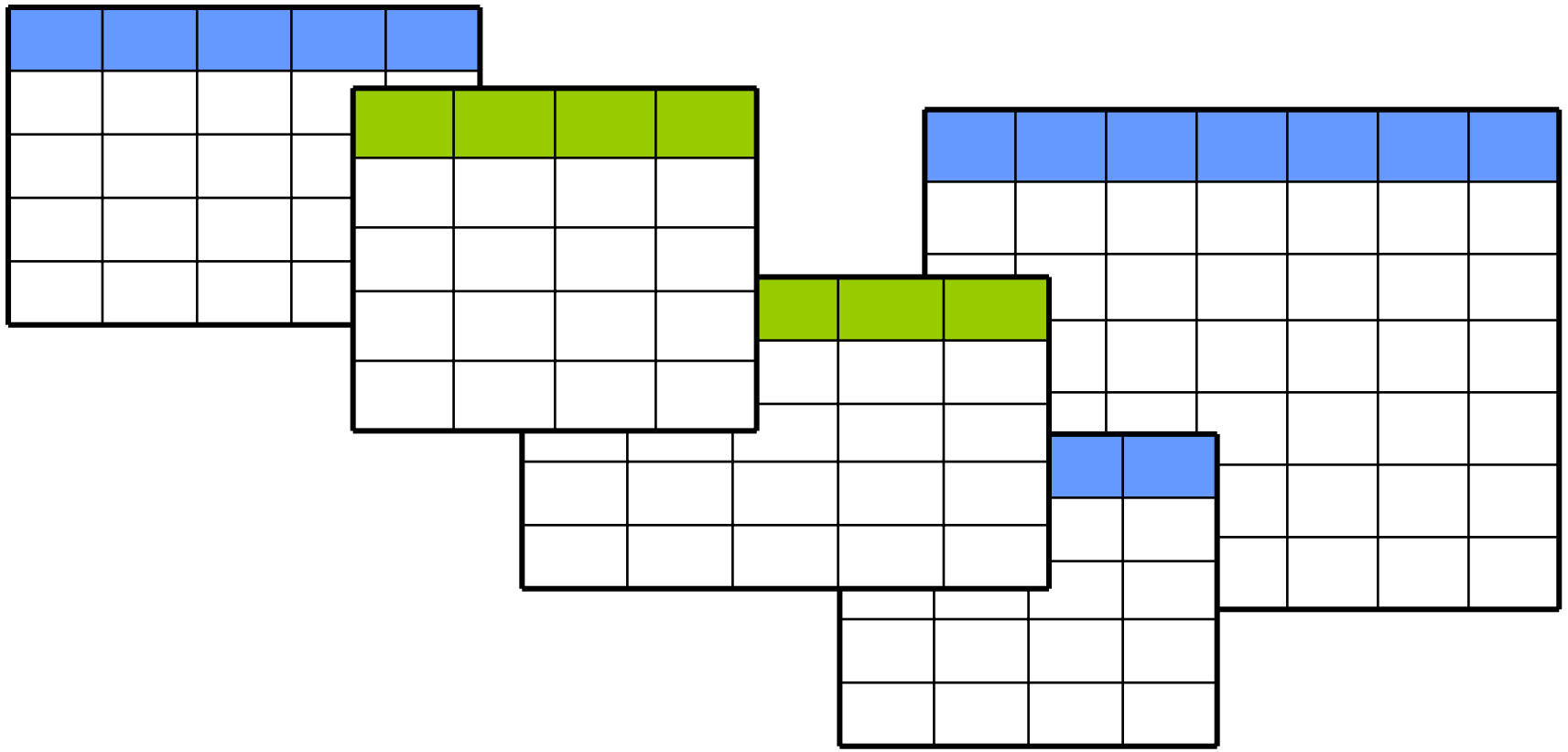
Automatic data transformation

- Data discovery
  - the data is profiled using profiling tools or sometimes using manually written profiling scripts to better understand the structure and characteristics of the data and decide how it needs to be transformed
  - similar to human guided data transformation
- Data mapping
  - define how individual fields are mapped, modified, joined, filtered, aggregated etc. to produce the final desired output
- Code generation
  - generating executable code (usually scripts) that will transform the data
- Code execution
  - the code is executed against the data to create the desired output
- Data review
  - ensuring the output data meets the transformation requirements

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At last: a tidy datasets

- Finally, we obtain a collection of tables



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Classification of datasets

- The obtained dataset can be classified with respect to different characteristics
  - number of tables
  - type of variables
    - we already saw this
  - number of variables
    - also called “dimension”
  - presence of dependent and independent variables
  - existence of relationships
    - flat and relational data
  - semantical classifications

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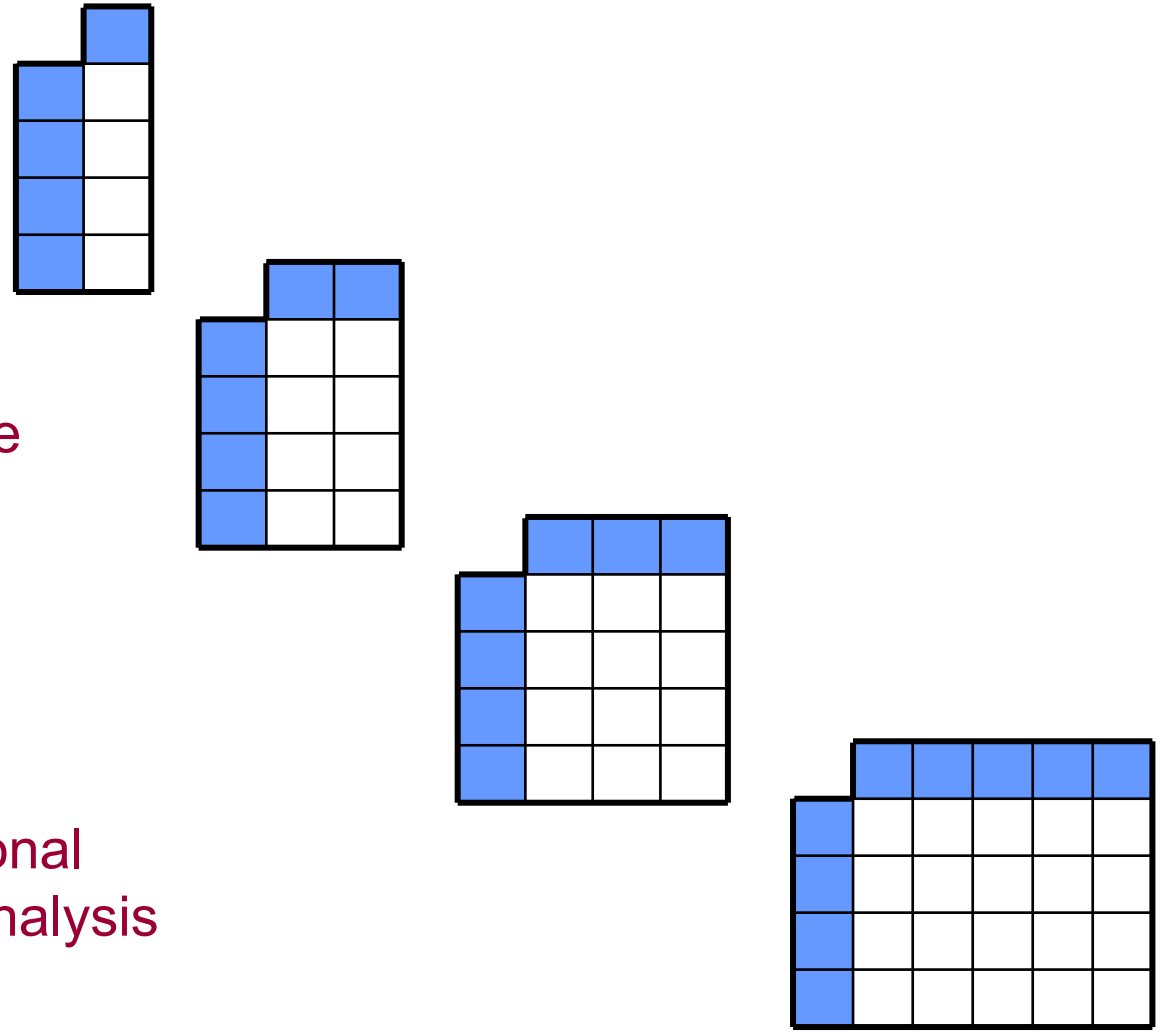
Multiple tables

- Define one table for each set of items
- Add a variable (a unique id) to the tables in order to allow you to quickly refer to an item (row) in the table
- When you save the tables as files
  - use a different file for each table
    - for example: avoid multiple spreadsheets in the same excel file
  - save variable names on the first line
    - use human readable names rather than encoded ones

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Dataset dimension

- The dimension of the data is the number of variables of each record
  - univariate data
    - one variable only
    - sometimes obtained by a projection from a richer dataset
  - bivariate data
    - two variables only
    - usually, one independent and the other dependent
  - trivariate data
    - three variables only
  - multivariate
    - more than three variables
    - or hypervariate, or multidimensional
    - the most difficult case for data analysis



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Dependent and independent variables

- Some variables may be viewed as dependent on the values of others
- Dependent variables are those that are supposed to be a consequence of independent variables
  - for the same combination of independent variables, you can have only a single value of each dependent variable

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Flat and relational datasets

- Flat dataset
  - only one type of object (and its values)
  - no relationships
  - can be further analyzed based on the number and type of values for each object
- Relational dataset
  - several types of objects, each one with its values
  - relationships between objects

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Relationships

- A relationship is a table that establishes a link among the data points of other tables
  - simplest case: binary relationship for a single data table

Id	Name	SSN	Age	Hair	Grade	…
P1	Mary	145	23	brown	B	…
P2	Jim	29				
P3	Sally	56				
P4	Mitch	82				
…		…				

Data table

Caller	Called	Time	Duration	…
P1	P2	9:15	0:30:24	…
P2	P1	23:17	0:02:35	…
P3	P2	23:47	0:3:00	…
P4	P3	23:51	0:25:33	…
	…	…	…	…

Relationship

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Relationships

- Types of relationships
  - one-to-one (binary relationship)
    - may be directed or undirected
    - the most common type of relationship
  - one-to-many
  - many-to-many
- Relationships may have attributes

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A semantic classification of datasets

- Special types of datasets based on the semantic content
  - geolocated/geospatial data
    - when some variables express geographic locations
  - grid/spatial fields
    - each datum is associated with a cell of a bi-dimensional or three-dimensional space
  - time series
    - when variables vary with respect to time (the only independent variable)

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Geolocated or geospatial data

- One, two, or three variables may be interpreted as geometric or geographic coordinates
  - Geolocation by nominal variables
    - Example: “Italy”, “France”, “Spain”
  - Geolocation by quantitative variables
    - Example: latitude and longitude
  - Geolocation by ordinal variables
    - Example: a grid with an implicit ordering of the cells

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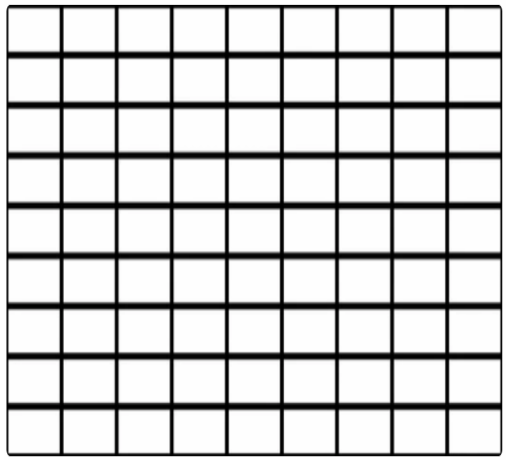
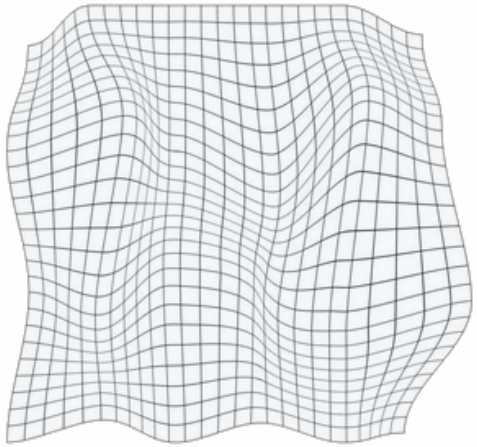
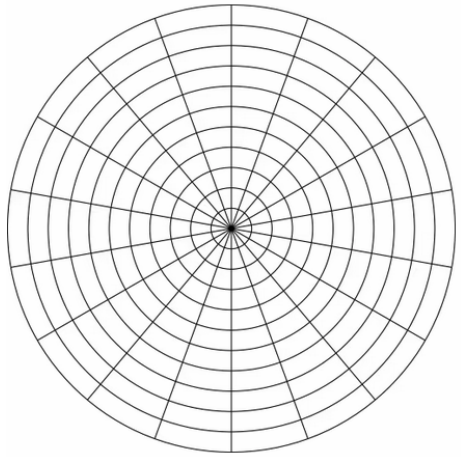
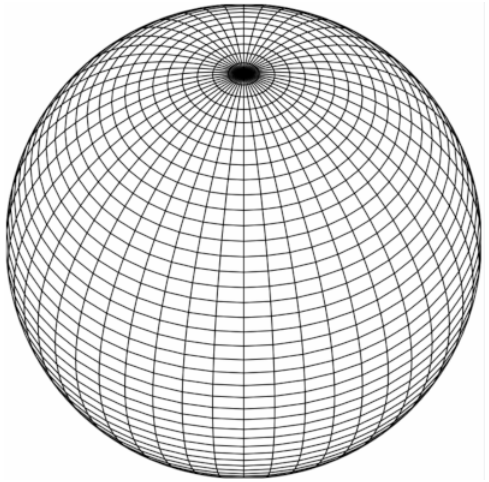
Metadata for geolocated or geospatial data

- Additional informations known in advance may include
  - the space geometry
    - Examples: the boundaries of the districts/region/countries that are named by the nominal variables
  - additional geometric objects
    - Examples: the green areas, the rivers, the mountains, etc.
  - additional nominal data
    - Examples: the name of a river, the name of a mountain, the name of a sea, etc.

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Grid or spatial fields

- Each data case has exactly two variables
  - the first variable is an ordinal that identifies a cell of a given space
    - this space is described by the associated metadata
  - the second variable provides a quantitative value

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Grid or spatial fields

- In practical cases the first value can be omitted replacing it with the order according to which the values are stored
  - this saves space when storing the dataset
  - metadata describe how the ordered values are mapped to the grid cells
  - in this case the dataset is actually monodimensional

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Time series

- One variabile provides a temporal indication
- Metadata describe
  - how to interpret the punctual data
    - an atomic value, a value referred to an interval, a cumulative value of all the past
  - what is the subdivision of the time
    - seconds, minutes, hours, days, etc.
  - extra temporal information available
    - day of the week, known events, etc.

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Bibliography and further readings

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